

Gravitational decoherence does not certify its mechanism: the Gaussian onset, CP-divisibility, and an operational undecidability theorem

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preprint, not peer reviewed, version 2026-06-14. Every load-bearing line is tagged by epistemic status: [THEOREM] = proved here (symbolic or high-precision numeric); [RESULT] = an established external or corpus result used as input; [OPEN] = an unsolved obligation. All numeric claims were verified at mpmath `dps 80` (commonly 100) and symbolically (sympy) where exact; the receipt script is `code/v6_pX_decoherence_undecidability_receipts.py`.

Abstract

A gravitationally induced loss of matter coherence is the most-discussed empirical window on the quantum-gravity interface, and a recurring claim is that its *shape* — most often a short-time **Gaussian onset** $| \langle T \rangle | = e^{-\frac{1}{2}(T/\tau_G)^2}$, as in finite-memory variants of the Diósi–Penrose proposal — diagnoses the underlying mechanism (non-Markovianity, state-dependence, or a genuinely quantum-gravitational record process). We prove the opposite, in a sharp and useful form: **the gravitational dephasing curve is mechanism-blind, and the structural distinction that would mark a quantum-gravitational record process is not a function of the reduced channel**. Three results, each computed at high precision. (i) The Gaussian onset is *not a signature*: a state-independent, irreversible record commitment (“seal”) firing at a ramping hazard $\langle t \rangle = a t$ reproduces it exactly ($| \langle T \rangle | = e^{-aT^2/2}$, residual 0), and a constant hazard gives the pure exponential semigroup — the most Markovian channel there is. (ii) Any state-independent irreversible seal is **CP-divisible**: $| \langle T \rangle | = e^{-\dots}$ is monotone, so it can *never* produce revivals; revivals occur exactly for Gaussian kernels whose running integral goes negative — the underdamped (still spectrally positive) kernel is the canonical case, with the boundary at $\dots = 3.644173671645632\dots$ — a non-Markovianity **orthogonal** to the structural axis. (iii) **Operational undecidability from the dephasing curve**: a continuous (Ornstein–Uhlenbeck) classical-noise ontology and a matched sparse-seal record ontology produce **bit-identical free-induction coherence** $| \langle T \rangle |$ for all T (the identity $C_{OU} = C_{seal}$ is sympy-exact; an independent numerical integration gives the machine-zero residual $\max | C_{OU} - C_{seal} | = 7.1 \times 10^{-12}$), so free-induction decay cannot separate them. Multi-time dynamical decoupling (echo/CPMG) *can* probe reversibility — it distinguishes a refocusable noise from an irreversible commitment — but the genuine quantum-gravitational content, whether the matter’s *closed-system* record process is *indivisible* in Barandes’ sense / *unrefinable* in the sealed-record sense, is **structural** (the off-diagonal support of the decoherence functional between sparse commitments) and is *invisible to passive reduced-channel measurement*, at every order, because a bath dilation is always divisible and a closed unitary system is indivisible yet a trivial channel

(an invasive protocol that reconstructs $\Gamma(\tau)$ would insert a seal at each conditioning step, and so cannot certify the unintervened process — §6, O7). We make the structural axis precise, exhibit the consistent-histories functor that turns “unrefinable indivisible” into a theorem at the kinematic level, and isolate the one open obligation (whether the gravitational record process is sparsely or continuously sealing). The widely used finite-memory Diósi–Penrose-family kernel is, on this analysis, *one illustrative CP-divisible member of the undecidable class rather than a discriminating prediction*. The practical upshot is a sharpened no-go: a free-induction onset-shape measurement, however precise, does not distinguish gravitational quantum decoherence from classical gravitational noise — the reversibility of the mechanism is probed only by multi-time decoupling, and the structural indivisibility that would mark a genuinely quantum-gravitational record process is reachable by no reduced-channel measurement at all.

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1. The question, and why the obvious answer is wrong

Suppose two branches of a massive body’s wavefunction carry distinct mass-density configurations, so that a relative gravitational self-energy ΔE_G accumulates a relative phase. A gravitationally induced decoherence model posits that this phase is degraded — the off-diagonal $\langle T \rangle$ of the which-path density matrix decays — and proposes a decay law. The Diósi–Penrose (DP) proposal gives a short-time **exponential** e^{-T/τ_G} with $\tau_G = \hbar/E_G$; finite-memory variants instead give a **Gaussian onset** $e^{-\frac{1}{2}(T/\tau_G)^2}$ crossing over to the DP slope at late times. A large experimental program — interferometric and (dominantly) non-interferometric — aims to measure this decay and thereby test “gravitational” collapse against quantum mechanics.

The temptation, repeatedly voiced, is to read the *shape* of the decay as a fingerprint of the mechanism: a Gaussian (rather than exponential) onset is taken as evidence of finite memory / non-Markovianity, and non-Markovianity in turn as a hint of a genuinely quantum-gravitational, non-classical record process. **Each link in that chain fails.** This paper proves three obstructions that, together, say the reduced channel is mechanism-blind, and then locates precisely where the genuine content does live.

The setup is the standard one. For a stationary Gaussian pure-dephasing channel,

$$C(T) = \langle |T| \rangle / \langle |0| \rangle = e^{-\langle T \rangle}, \quad \langle T \rangle = (1/\tau^2) \int_0^T (T-s) K(s) ds,$$

with $K(s) = E_G(0) E_G(s)$ the noise autocorrelation, and the instantaneous dephasing rate is $\tau(T) = 2 \langle T \rangle = (2/\tau^2) \int_0^T K(s) ds$. The DP-family kernel is the Ornstein–Uhlenbeck (exponential) form $K(s) = \tau^2 e^{-|s|/\tau_c}$; we return to it in §5. The point of departure is that the *closed-system* record process underlying such a channel — the sealed-record ontology in which a “division event” commits a which-path record and the gravitational holonomy between events is a coherent, uncommitted phase — is a *different* object from the noise $E_G(\tau)$, and the two can be told apart, if at all, only structurally.

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2. Three inequivalent senses of "Markovian", and where the fingerprint lives

The word "Markovian" collides three inequivalent notions, and a given model's status flips between them. Naming them is what dissolves the false chain of §1.

1. **Constant-rate semigroup** (time-homogeneous Lindblad). A finite-memory channel is *not* this: its rate $\Gamma(T)$ is time-dependent — that time-dependence *is* the Gaussian onset. This is the only sense in which "the Gaussian-onset channel is non-Markovian" is unambiguously true, and it is the weakest sense.
2. **CP-divisibility** (the RHP/BLP sense: a time-dependent but pointwise-positive rate, no information backflow). A channel is CP-divisible **iff** $\int_0^T K(s) ds \geq 0$ for all T — equivalently $\Gamma(T) \geq 0$, $|C(T)|$ monotone non-increasing. This is *strictly stronger* than the kernel being positive-type (spectral density $S(\omega) \geq 0$): positivity of S is a statement about the *full-line* Fourier transform of K , while CP-divisibility is governed by the sign of the *finite running integral* $\int_0^T K$.
3. **Markov embeddability** (dilation to a memoryless process on a larger space). By Mori–Zwanzig essentially every physical channel has one; the OU kernel is a 1-D Ornstein–Uhlenbeck embedding, the underdamped kernel of §3 a 2-D one.

These three axes are **orthogonal**, and all corners are populated:

- A closed unitary qubit is **indivisible** in Barandes' structural sense (§4) yet, as a channel, *trivial* (no decoherence at all) — structural indivisibility with no operational signature.
- An underdamped Gaussian noise is **non-CP-divisible** (it revives, §3.4) yet **Markov-embeddable** and *not* structurally indivisible — an operational signature with no structural content.

So **CP-divisibility and structural indivisibility are independent axes; non-CP-divisibility is neither necessary nor sufficient for the structural barrier**. The whole of §3 is the consequence of taking this seriously: the things one can measure on $| \rangle$ (rate time-dependence, revivals) live on axes 1–2, while the quantum-gravitational question of interest lives on a fourth, structural, axis that §4 makes precise.

3. The reduced channel is mechanism-blind: three theorems

We take the closed-system object seriously: a which-path qubit whose branches carry distinct gravitational records, with a deterministic holonomy $\Phi(t) = (1/\hbar) \int_0^t \Delta E_G dt$ between **division events** (seals), and a **state-independent** irreversible seal firing at hazard $\Gamma(t) \geq 0$. A seal commits the which-path record; irreversibility means the branch-record overlap $\langle E(t) | E(t) \rangle$ is a non-increasing contraction. Then

$$\langle E(t) | E(t) \rangle = e^{-\int_0^t \Gamma(t) dt} \langle E(0) | E(0) \rangle = e^{-\int_0^t \Gamma(t) dt} \langle E(0) | E(0) \rangle.$$

3.1 The Gaussian onset is not a signature [THEOREM]

A **ramping** hazard $\lambda(t) = a t$ gives $\lambda = aT^2/2$, hence

$$|\lambda(T)| = e^{-aT^2/2},$$

which is *exactly* the Gaussian onset with $a = 1/\gamma^2$ (symbolic identity; numeric residual 0.0 at every T on the grid 0.1,...,7, $\text{dps} = 100$; e.g. $e^{-a \cdot 3.5^2/2} = 2.1874911181828851 \times 10^{-3}$, residual 0). The initial slope $d/dT \ln|\lambda|$ vanishes at $T = 0$ — the quadratic (Zeno-like) onset — with no fine-tuning. A *constant* hazard λ instead gives the pure exponential semigroup $e^{-\lambda T}$, the most Markovian channel of all. **The onset shape is set by the time-profile of a state-independent commitment rate, and carries no information about whether the underlying process is classical noise or a quantum record process.** The Gaussian is the ramp; the exponential is the constant; neither is a fingerprint of anything but $\lambda(t)$.

3.2 State-independent irreversible sealing is CP-divisible — it cannot revive [THEOREM]

Because $\lambda(t) \geq 0$, the coherence $|\lambda(T)| = e^{-\lambda T}$ is **monotone non-increasing**: the channel is CP-divisible, with no information backflow and no revivals, for *every* state-independent irreversible seal. For the DP-family OU kernel this is explicit: $K = \gamma^2 \gamma_c (1 - e^{-T/\gamma_c})$, so $\lambda(T) = (\gamma^2 \gamma_c / 2) (1 - e^{-T/\gamma_c}) \geq 0$ (the rate minimum over a wide grid is 0.0, attained at $T = 0$). A non-monotone $|\lambda|$ would require $E|E$ to *grow* — the record returning which-path information, i.e. reversible echo / an information-preserving bath (by definition *not* a committed record), or a seal reading the coherence to engineer refocusing (the forbidden nonlinear state-dependence; §6, O2). Hence the dichotomy, with no middle ground:

state-independent + irreversible **monotone, CP-divisible, no revival; revivals** **no committed record, or state-dependence.**

3.3 Operational undecidability: continuous noise and sparse sealing give identical free-induction decay to all orders (in T) [THEOREM]

The decisive obstruction. Take the DP-family OU dephasing curve $C_{\text{OU}}(T) = e^{-\lambda(T)}$. Its CP-divisibility (§3.2) means its equivalent instantaneous hazard $\lambda_{\text{OU}}(t) = \lambda(t) = (\gamma^2 \gamma_c / 2) (1 - e^{-t/\gamma_c})$ is globally ≥ 0 — hence λ_{OU} is *itself* a legitimate state-independent irreversible-seal hazard. A sparse-seal record process with that hazard therefore reproduces the continuous-noise coherence **to all orders**:

$$\int_0^T \lambda_{\text{OU}}(t) dt = \lambda(T) \text{ identically (sympy-exact), so } C_{\text{seal}}(T) = e^{-\int_0^T \lambda_{\text{OU}}(t) dt} = e^{-\lambda(T)} = C_{\text{OU}}(T) \text{ for all } T.$$

Numerically, integrating λ_{OU} independently and comparing, $\max |C_{\text{OU}}(T) - C_{\text{seal}}(T)| = 7.1 \times 10^{-12}$ across $T = 0.1, \dots, 10$ at $\text{dps} = 100$ (machine zero). The "order of first disagreement" between a *truncated* seal model and OU ($O(T^3)$ for a ramp, $O(T)$ for an n -term hazard) is a free truncation knob, not physics; at full resolution the two coincide exactly. So **free-induction decay does not separate the continuous-noise ontology (gravity as a classical stochastic field) from the matched sparse-seal ontology (gravity as a discrete record process)** — the single-time curve, and the Gaussian-equivalent two-time function it implies, are shared (the genuine multi-time statistics differ, as the next paragraph notes).

This is a statement about *free induction*, and that is where to be careful. A multi-time **dynamical-decoupling** sequence (spin echo, CPMG) is *not* a function of $| \langle T \rangle |$ alone: it refocuses reversible phase accumulation, so it *can* separate a refocusable noise (whose coherence echo restores) from an irreversible commitment (whose committed records echo cannot undo), and it is sensitive to the non-Gaussian higher cumulants by which the two models differ. So *reversibility is an accessible discriminant*. What is *not* accessible at any order is the **structural** axis of §4 — whether the matter’s *closed-system* record process is indivisible: a bath dilation is always divisible (§4.1) and a closed unitary system is indivisible yet a trivial channel (§2), so no *passive* reduced-channel measurement, free-induction or multi-time, is a function of it; an *invasive* protocol that reconstructs $\Gamma(t)$ requires intermediate conditioning measurements, but each such measurement inserts a seal, so it cannot certify the *unintervened* process’s indivisibility (the do-delete obstruction; §6, O7). The dephasing curve hides the mechanism; the whole channel hides the structure.

3.4 What a revival would, and would not, buy you [THEOREM]

Revivals (operational non-Markovianity, axis 2) are reachable — but only off the committed-seal class, and they still do not reach the structural axis. The equally positive-type **underdamped** kernel $K(s) = \frac{1}{2} e^{-|s|/\tau} \cos(\omega s)$ (a Lorentzian-pair spectrum, $S(\omega) \geq 0$) has running integral

$$K = \frac{1}{2} \cdot [1 + e^{-T/\tau} (\sin \omega T - \cos \omega T)] / (1 + (\omega \tau)^2),$$

whose minimum over T first dips below zero at

$$(\omega \tau)_* = 3.644173671645632136171\dots$$

(the value where the running integral becomes tangent to zero; $\text{dps} = 80$), and is negative for $\omega \tau > (\omega \tau)_*$ — verified revivals at $\omega \tau = 5$ ($\omega_{\min} = -0.0729$ at $T_* = 0.9425$) and $\omega \tau = 10$ ($\omega_{\min} = -0.1038$ at $T_* = 0.4712$). So revivals are a property of the *kernel’s oscillation*, available within Gaussian noise, and they certify only non-CP-divisibility. A sharper, genuinely discrete fingerprint is the bimodal limit $E_G = \pm E$, giving $C(T) = \cos(E T/\tau)$ (symbolic; max residual 5.5×10^{-12}) — a hard which-path oscillation with $C(\omega \tau) = -1$, distinct from the smooth Gaussian echo. This is the most diagnostic *observable* of discreteness, but it is a **diagnostic, not a definition**: it is neither necessary nor sufficient for the structural barrier (§2), and §3.2 shows a committed seal cannot produce it at all.

4. Where the content actually lives: the structural (indivisibility) axis

The quantum-gravitational question that the reduced channel cannot answer is whether the matter’s gravitational record process is **indivisible**.

4.1 The barrier in two languages [RESULT: Barandes; sealed-record program]

Barandes specifies a process by its one-time-conditioned transition matrices $\Gamma(t)_{ji} = P(j \mid i \text{ at } t \mid 0)$. The process is **divisible** (his Markovian) iff $\Gamma(t, t) = I$.

$= \Gamma(\mathbf{t}, \mathbf{t})\Gamma(\mathbf{t}, \mathbf{t})$ for *every* intermediate $\mathbf{t}$, and **indivisible** iff composition fails except at sparse **division events**. The stochastic–quantum correspondence makes the obstruction explicit, $\Gamma(\mathbf{t})_{\mathbf{j}\mathbf{i}} = |\mathbf{U}(\mathbf{t})_{\mathbf{j}\mathbf{i}}|^2$:

$$[\Gamma(\mathbf{t})\Gamma(\mathbf{t})]_{\mathbf{j}\mathbf{i}} = \sum_{\mathbf{k}} |\mathbf{U}(\mathbf{t})_{\mathbf{j}\mathbf{k}}|^2 |\mathbf{U}(\mathbf{t})_{\mathbf{k}\mathbf{i}}|^2 \text{ (sum of path probabilities),}$$

$$\Gamma(\mathbf{t}\mathbf{t})_{\mathbf{j}\mathbf{i}} = |\sum_{\mathbf{k}} \mathbf{U}(\mathbf{t})_{\mathbf{j}\mathbf{k}} \mathbf{U}(\mathbf{t})_{\mathbf{k}\mathbf{i}}|^2 \text{ (probability of the path sum),}$$

and the gap between them is the **interference cross-term**. Divisibility holds exactly where that cross-term vanishes — a division event. In the sealed-record language the same line is the **unrefinability of the seal order**: between commitments the system carries a reversible holonomy (a closed phase, no committed value); one cannot insert an intermediate conditioning record without *sealing* it, and sealing destroys the holonomy and changes the process. *Refine insert an intermediate conditioning record compose the transition matrices Chapman–Kolmogorov*, so the two barriers are the same barrier. Two features matter here: (i) indivisibility is a property of the *closed system’s own* transition law — a bath dilation always gives a *divisible* embedding, which is exactly why a noise model is structurally silent; (ii) it is a strictly *different* axis from open-system CP-divisibility (the closed unitary qubit is indivisible yet a trivial channel).

4.2 The functor: “unrefinable indivisible” is a theorem at the kinematic level [THEOREM, kinematic; gravitational realization OPEN]

The bridge is the Gell-Mann–Hartle decoherence functional. A sealed-record history is a finite poset of seals $\mathbf{e}_{\mathbf{k}}$ with a unitary holonomy $\mathbf{U}_{\mathbf{k}}$ on each inter-seal interval and a pointer-basis commitment $\mathbf{P}_{\mathbf{a}}$ at each seal. Define

$$\mathbf{D}(\mathbf{e}, \mathbf{e}) = \text{Tr}[\mathbf{P}_{\mathbf{a}_n} \mathbf{U}_{\mathbf{a}_{n-1}\mathbf{a}_n} \mathbf{P}_{\mathbf{a}_{n-1}} \mathbf{U}_{\mathbf{a}_{n-2}\mathbf{a}_{n-1}} \mathbf{P}_{\mathbf{a}_{n-2}} \dots \mathbf{U}_{\mathbf{a}_1\mathbf{a}_2} \mathbf{P}_{\mathbf{a}_1}]$$

on coarse-grained pointer histories, with the projectors evolved by the inter-seal holonomies, and read the one-time matrix from its diagonal, $\Gamma(\mathbf{t})_{\mathbf{j}\mathbf{i}} = \mathbf{D}(\mathbf{j}; \mathbf{i})$. Then the Tier-1 biconditional *is* the consistent-histories consistency condition:

Chapman–Kolmogorov holds across $\mathbf{t}$ the off-diagonal (interference) part of $\mathbf{D}$ vanishes at $\mathbf{t}$ (medium decoherence) $\mathbf{t}$ is a seal.

Forward: a projective commitment diagonalizes $\mathbf{D}$ at $\mathbf{t}$, which is exactly the probability sum rule (Chapman–Kolmogorov) for the pointer histories. Reverse: if Γ composes across $\mathbf{t}$ for *all* boundary conditions, the interference terms must vanish at $\mathbf{t}$, i.e. the record is committed. So **unrefinable indivisible is a theorem of consistency** — *for a chosen projective seal-history model*. What is **not** thereby settled, and is the one open obligation, is whether the *gravitational* record process supplies these very seals; that is §4.3, not something the functor decides.

4.3 The one open question, sharply posed [OPEN]

Is gravitational record commitment sparse or continuous?

If the gravitational records decohere the which-path *continuously* (medium decoherence at all times — the continuous-sealing limit), $\mathbf{D}$ is diagonal everywhere, the seal order is fully refinable, and the process is **divisible** (classical), with the dephasing curve of §5 its complete content. Genuine indivisibility requires *intervals on which $\mathbf{D}$ is non-diagonal* — interference surviving between sparse gravitational seals. This is a computation on the off-diagonal support of $\mathbf{D}$, not

on $| \quad |$, and by §3 it cannot be read off any coherence-decay measurement. It is the correct target, and it is open.

5. The Diósi–Penrose-family kernel, in its true status

The widely used finite-memory channel is a fully legitimate, empirically anchored member of the undecidable class — and nothing more. With the OU kernel,

$$(T) = (\sigma^2 \tau_c^2 / 2) (T / \tau_c - 1 + e^{-T / \tau_c}),$$

which is quadratic at short times ($(\sigma^2 / 2 \tau_c^2) T^2$, the Gaussian onset) and crosses over to the linear DP slope at $T = \tau_c$. Matching the onset to the DP scale $\tau_G = \tau_c / E_G$ and the curvature condition $(\tau_c) = 1$ fixes the single internal time,

$$\tau_c = \sqrt{e} \cdot \tau_G = 1.6487212707001281468486507878141635716537761007101 \cdot \tau_G$$

(the fixed point $(\tau_c / \tau_G)^2 = e$ is exact; residual $|(\tau_c) - 1| = 7.1 \times 10^{-12}$ at $\text{dps} = 100$). The energy scale E_G is inherited from the DP geometric self-energy and is *not* derived here; the model is **rate-robust, not fit-free**. Its empirical content — the quadratic onset against the DP exponential, the E_G -scaling, the crossover, the small higher-cumulant (non-Gaussian) correction — is exactly what a free-induction experiment can access, and §3.3 is the statement that *the free-induction content is shared with the classical-noise ontology*. The kernel is therefore the program’s **dense-seal Gaussian shadow** — it recovers the phenomenology and loses no empirical ground — and is, correctly, one illustrative CP-divisible member of the undecidable class, not the discriminating prediction the onset shape is often read as.

[**RESULT: DP-family phenomenology, ref. [11,12].**] In the large-object, large-displacement regime relevant to the experiments [11, 12], the finite-memory modification constrains only the model’s regularization length R_0 (per-nucleon emission $1/R_0^3$) and decouples from the collective decoherence the interferometer measures ($\propto \tau_G^2 / R, R_0$ -independent); it does not *worsen* the radiation/heating bounds and in fact *suppresses* high-frequency (keV) emission. So the kernel survives the laboratory bounds — which is exactly why it is a viable member of the undecidable class, not a falsified one.

6. The observability ledger: what a measurement can and cannot decide

A coherence-decay experiment is a measurement of $C(T) = e^{-C(T)}$ and its statistics. The following constrain what such a measurement can certify; each is a [RESULT] (corpus or external) used as a constraint on the *interpretation*, not a new claim.

#	Constraint	Consequence for a decoherence test
O1	The Gaussian onset is reproduced by a state-independent ramp (§3.1)	The onset stage certifies neither non-Markovianity in any strong sense nor a quantum-gravitational mechanism.
O2	State-independent irreversible scaling is CP-divisible (§3.2)	A revival, if seen, rules out a committed-record interpretation of that channel (forces reversibility or state-dependence); its absence certifies nothing.
O3	OU-noise and matched sparse-seal share the free-induction curve (§3.3)	Free-induction decay does not distinguish classical gravitational noise from a discrete record process; multi-time decoupling probes reversibility, but the structural axis (O4–O6) is invisible to the whole channel.
O4	Both dilations are always divisible (§4.1)	Fitting an open-system (master-equation) model can never exhibit the closed-system indivisibility, by construction.
O5	A closed unitary system is indivisible yet a trivial channel (§2)	The structural axis is not a channel property; it cannot be the output of any reduced-channel tomography.
O6	Bell non-evision [RESULT: corpus]	The memory may not be used to signal or to evade Bell; the channel’s non-Markovianity relocates nonlocality into memory, it does not remove it.
O7	Reconstructing $T(C)$ needs intermediate conditioning [RESULT: do-delete]	An invasive multi-time protocol can reconstruct Blumaker’s $T(C)$, but each conditioning measurement inserts a seal (commits a record), so it cannot certify the <i>unintervened</i> process’s indivisibility — the intervention is not an observational statistic.

The reading is uniform: *every* discriminating question about the mechanism lives on the structural axis (§4), and the structural axis is provably not a function of the reduced channel. The

one thing a measurement *can* do is constrain E_G and the onset time within the shared phenomenology — useful, but mechanism-blind.

7. What this paper does and does not claim

It **claims**, with verified mathematics: the Gaussian onset is reproduced exactly by a state-independent ramping seal and is not a mechanism signature (§3.1); any state-independent irreversible seal is CP-divisible and cannot revive (§3.2); continuous classical noise and sparse sealing produce identical reduced coherence to all orders, so the mechanism is operationally undecidable from the channel (§3.3); revivals are a separate, orthogonal axis reachable only off the committed-seal class (§3.4); and the structural (indivisibility) question is well-posed, reduces at the kinematic level to a consistency theorem via the decoherence functional (§4.2), and is invisible to $| \cdot |$.

It does **not** claim: that the sealed-record/indivisible ontology is the *unique* mechanism (the undecidability cuts both ways — it equally forbids certifying *classical* noise); that the gravitational record process *is* indivisible (whether it is sparsely or continuously sealing is [OPEN], §4.3); that any observable non-Markovian fingerprint of indivisibility exists (§3.2 rules it out for the committed-record class); a derivation of general relativity, a graviton, or the Born rule; or a solution of the relativistic-collapse (Tomonaga–Schwinger) and covariant-localization (Hegerfeldt) obstructions, which bound any concrete realization. The finite-memory DP-family kernel is presented as one CP-divisible member of the undecidable class, not as a prediction.

The honest one-line summary: **gravitational decoherence has a shape, and the shape does not know its own cause; the cause is structural, and the structure does not show up in the decay curve.**

8. Precision discipline

Every quantity was computed in mpmath at `dps = 80` (commonly 100), with sympy used for the exact identities (the Gaussian-onset integral, the OU running integral, the $_{OU} =$ all-orders identity, the underdamped running integral, the bimodal cosine). Float-safe small quantities are stated as such. The receipt script `code/v6_pX_decoherence_undecidability_receipts.py` reproduces every number in the canonical output block: the Gaussian onset (residual 0), the CP-divisible OU rate ($_{min} = 0$), $\sqrt{e} = 1.6487212707\dots$ (50 digits), the revival threshold ($_{*} = 3.644173671645632\dots$), the all-orders identity $\max|C_{OU} - C_{seal}| = 7.1 \times 10^{-12}$, and the bimodal cosine ($\max$ residual 5.5×10^{-12}). Never float64 for the kernel running-integrals or the all-orders comparison, where cancellation makes double precision unreliable.

References

External.

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Companion papers.

[V] *Quantum theory from sealed records I–II*, companion papers ([paper-Va-foundations-1](#), [paper-Vb-foundations-2](#)) — axioms R, S, C; the $q = 2$ weight calculus. [XI] *Emergent Einstein equations without an emergent Newton constant*, companion paper ([paper-XI-sealed-record-gravity-no-go](#)) — the gravitational equation of state and the scale no-go; the indivisibility question of §4 is the observability counterpart of its open covariance frontier.